

An Analysis of DenseNet

DenseNet

DenseNet -- Part 1

Projection connection If the function F is of type $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$ where $n \neq m$, then $F(x) + x$ is undefined. To handle this special case, a projection connection is used:

DenseNet -- Part 2

Bottleneck block A bottleneck block consists of three sequential convolutional layers and a residual connection. The first layer in this block is a 1×1 convolution for dimension reduction (e.g., to $1/2$ of the input dimension); the second layer performs a 3×3 convolution; the last layer is another 1×1 convolution for dimension restoration. The models of ResNet-50, ResNet-101, and ResNet-152 are all based on bottleneck blocks.

DenseNet -- Part 3

During the early days of deep learning, there were attempts to train increasingly deep models. Notable examples included the AlexNet (2012), which had 8 layers, and the VGG-19 (2014), which had 19 layers. However, stacking too many layers led to a steep reduction in training accuracy, known as the "degradation" problem. In theory, adding additional layers to deepen a network should not result in a higher training loss, but this is what happened with VGGNet. If the extra layers can be set as identity mappings, however, then the deeper network would represent the same function as its shallower counterpart. There is some evidence that the optimizer is not able to approach identity mappings for the parameterized layers, and the benefit of residual connections was to allow identity mappings by default. In 2014, the state of the art was training deep neural networks with 20 to 30 layers. The research team for ResNet attempted to train deeper ones by empirically testing various methods for training deeper networks, until they came upon the ResNet architecture.

DenseNet -- Part 4

This formulation suggests that the gradient computation of a shallower layer, $\frac{\partial E}{\partial x_i}$, always has a later term $\frac{\partial E}{\partial x_L}$ that is directly added. Even if the gradients of the $F(x_i)$ terms are small, the total gradient $\frac{\partial E}{\partial x_i}$ resists vanishing due to the added term $\frac{\partial E}{\partial x_L}$.

DenseNet -- Part 5

$$\begin{aligned} x_i + 2 &= F(x_i + 1) + x_i + 1 = F(x_i + 1) + F(x_i) + x_i \\ &\quad \begin{aligned} &\begin{aligned} &x_{\ell} + 2 &= F(x_{\ell} + 1) + F(x_{\ell}) + x_{\ell} \\ &+ 1 &= F(x_{\ell} + 1) + F(x_{\ell}) + x_{\ell} \end{aligned} \end{aligned} \end{aligned}$$

DenseNet -- Part 6

Subsequent work Wide Residual Network (2016) found that using more channels and fewer layers than the original ResNet improves performance and GPU-computational efficiency, and that a block with two 3×3 convolutions is superior to other configurations of convolution blocks. DenseNet (2016) connects the output of each layer to the input to each subsequent layer:

DenseNet -- Part 7

During the late 1980s, "skip-layer" connections were sometimes used in neural networks. Examples include: Lang and Witbrock (1988) trained a fully connected feedforward network where each layer skip-connects to all subsequent layers, like the later DenseNet (2016). In this work, the residual connection was the form $x \mapsto F(x) + P(x)$, where P is a randomly-initialized projection connection. They termed it a "short-cut connection". An early neural language model used residual connections and named them "direct connections".

DenseNet -- Part 8

Basic block A basic block is the simplest building block studied in the original ResNet. This block consists of two sequential 3×3 convolutional layers and a residual connection. The input and output dimensions of both layers are equal.

DenseNet -- Part 9

Backward propagation The residual learning formulation provides the added benefit of mitigating the vanishing gradient problem to some extent. However, it is crucial to acknowledge that the vanishing gradient issue is not the root cause of the degradation problem, which is tackled through the use of normalization. To observe the effect of residual blocks on backpropagation, consider the partial derivative of a loss function E with respect to some residual block input x_{ℓ} . Using the equation above from forward propagation for a later residual block $L > \ell$:

DenseNet -- Part 10

Previous work Residual connections were noticed in neuroanatomy, such as Lorente de No (1938). McCulloch and

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Pitts (1943) proposed artificial neural networks and considered those with residual connections. In 1961, Frank Rosenblatt described a three-layer multilayer perceptron (MLP) model with skip connections. The model was referred to as a "cross-coupled system", and the skip connections were forms of cross-coupled connections.

DenseNet -- Part 11

where P is typically a linear projection, defined by $P(x) = Mx$ where M is a $m \times n$ matrix. The matrix is trained via backpropagation, as is any other parameter of the model.

DenseNet -- Part 12

where L is the index of a residual block and ℓ is the index of some earlier block. This formulation suggests that there is always a signal that is directly sent from a shallower block ℓ to a deeper block L .

DenseNet -- Part 13

Pre-activation block The pre-activation residual block applies activation functions before applying the residual function F . Formally, the computation of a pre-activation residual block can be written as:

DenseNet -- Part 14

Residual connection In a multilayer neural network model, consider a (non-residual) subnetwork with a certain number of stacked layers (e.g., 2 or 3). Let $H(x; \alpha)$ denote the subnetwork. Suppose H^* is the desired optimal output of this subnetwork. Residual learning simply adds x directly to the output, such that the optimal learned output now becomes $H^* - x$, which is interpreted as a "residual" with respect to x . The operation of "adding x " is implemented via a "skip connection" that performs an identity mapping to connect the input of the subnetwork with its output. This connection is referred to as a "residual connection" in later work. Let $F(x; \alpha) = H(x; \alpha) + x$. The function F is often represented by matrix multiplication interlaced with activation functions and normalization operations (e.g., batch normalization or layer normalization). As a whole, one of these subnetworks is referred to as a "residual block". A deep residual network is constructed by simply stacking these

blocks. Long short-term memory (LSTM) has a memory mechanism that serves as a residual connection. In an LSTM without a forget gate, an input x_t is processed by a function F and added to a memory cell c_t , resulting in $c_{t+1} = c_t + F(x_t)$. An LSTM with a forget gate essentially functions as a highway network. To stabilize the variance of the layers' inputs, it is recommended to replace the residual connections $x + f(x)$ with $x/L + f(x)$, where L is the total number of residual layers.